BANGALORE CLIMATE ANALYTICS SYSTEM

Complete Technical Analysis & Project Documentation

Advanced Climate Analysis | Anomaly Detection | Forecasting System

Project Title	Bangalore Climate Analytics System
Version	Final (v2.0)
Technology Stack	Python 3.10 | Flask | SARIMA | SQLite
Data Coverage	1951 – 2024 (73 Years | 882 Months)
Report Date	16 June 2026

1. Executive Summary

The Bangalore Climate Analytics System is a full-stack web application built using Python (Flask) that provides scientific climate analysis, anomaly detection, and 12-month forecasting for Bangalore, India. The system is backed by 73 years of daily meteorological data (1951–2024) sourced from the India Meteorological Department (IMD), covering rainfall, maximum temperature, minimum temperature, and relative humidity.

The application uses the World Meteorological Organization (WMO) Standardized Precipitation Index (SPI) methodology (WMO-No. 1090, 2012) to compute a composite Climate Extreme Score (CES) that quantifies the overall climate extremeness for any given month. SARIMA time-series models forecast future climate variables, and the 12-month CES forecast is dynamically driven by the user's current input conditions.

Key Features at a Glance

- 73-year historical climate database (882 monthly records)
- Z-score based anomaly detection for Rainfall, Temperature, and Humidity
- Composite Climate Extreme Score (CES) using WMO SPI percentile thresholds
- SARIMA forecasting for 12-month ahead predictions
- User-input-driven CES forecast: extreme inputs produce risk alerts in early forecast months
- Interactive web dashboard with user authentication (Flask-Login + SQLite)
- Explainable AI (XAI) page for scientific transparency
- 15+ analytical charts generated automatically
- Period, decade, and seasonal comparative analysis
- REST API endpoint (/api/forecast) for programmatic access

2. Project Architecture

2.1 Directory Structure

The project is organized as a standard Flask application with dedicated modules for each concern:

Path / File	Description
`bangalore_climate_app/`	Root application directory
`app.py`	Main Flask application (1,382 lines) — all routes, logic, helpers
`run_app.py`	Application launcher with startup checks
`data.csv`	Raw daily climate dataset (1951–2024, ~1.18 MB, ~27,000 rows)
`database.db`	SQLite database — users, prediction history
`requirements.txt`	Python package dependencies
`saved_models/`	Trained SARIMA model files (.pkl) — Rainfall, Temp_mean, Humidity
`utils/`	Core scientific utility modules
`preprocessing.py`	Data loading, cleaning, monthly aggregation
`anomaly_detection.py`	Z-score anomaly detection, baseline computation
`climate_score.py`	CES computation, trend analysis, visualization
`forecasting.py`	SARIMA training, forecasting, user-driven CES blending
`database.py`	SQLite ORM — user registration, authentication, history
`train_models.py`	Model training pipeline wrapper
`static/images/`	Generated forecast charts (result page)
`static/analysis/`	17 pre-generated analytical charts (XAI/analysis pages)
`templates/`	Jinja2 HTML templates (index, login, predict, result, analysis, xai)
`outputs/plots/`	Backup plot storage
`Backend/`	Standalone testing and experimentation scripts

2.2 Technology Stack

Layer	Technology	Version	Purpose
Backend Framework	Flask	2.3.3	Web server, routing, session management
Authentication	Flask-Login	0.6.2	User sessions, login protection
Data Processing	Pandas	2.0.3	Data wrangling, aggregation
Numerical Computing	NumPy	1.24.3	Array operations, Z-score computation
Machine Learning	Scikit-Learn	1.3.0	MAE, RMSE, R^2 metrics
Time Series	Statsmodels	0.14.0	SARIMA model fitting & forecasting
Auto ARIMA	pmdarima	2.0.4	Automatic order selection
Statistical Tests	SciPy	1.11.1	ADF test, linear regression
Visualization	Matplotlib	3.7.2	All charts and plots
Interactive Plots	Plotly	5.15.0	Interactive chart components
Model Persistence	Joblib	1.3.2	Save/load SARIMA .pkl models
Database	SQLite (built-in)	—	User accounts, prediction history
Frontend	HTML5 + CSS3 + JS	—	Jinja2 templates, custom styles
Server OS	Windows / Linux	—	Cross-platform deployment

3. Data Pipeline

3.1 Raw Dataset

The application uses a daily climate dataset for Bangalore (Kempegowda International Airport / HAL weather station) covering January 1951 to June 2024 — a span of 73 years. The raw CSV file (data.csv) contains approximately 27,000 daily records.

Column	Description	Unit	Notes
Date	Observation date	YYYY-MM-DD	Parsed as datetime
Rainfall	Daily rainfall accumulation	mm	'Tr'/'Trace' → 0.0
Temp_Max	Daily maximum temperature	°C	Used in Temp_mean
Temp_Min	Daily minimum temperature	°C	Used in Temp_mean
Humidity	Relative humidity	%	Daily average

3.2 Preprocessing Steps (preprocessing.py)

- Column standardization: rename raw headers to [Date, Rainfall, Temp_Max, Temp_Min, Humidity]
- Date parsing: pd.to_datetime() with error coercion; invalid dates dropped
- Trace rainfall handling: 'Tr' and 'Trace' strings replaced with 0.0 before numeric conversion
- Numeric coercion: pd.to_numeric(errors='coerce') on all measurement columns
- Duplicate removal: drop_duplicates() on Date column
- Missing value imputation: linear interpolation → forward fill → backward fill
- Derived feature: Temp_mean = (Temp_Max + Temp_Min) / 2
- Time features: Year and Month extracted from Date
- Filtering: records limited to 1951–2024 for analysis consistency

3.3 Monthly Aggregation

Daily data is aggregated to monthly frequency for all modeling and analysis:

Variable	Aggregation Method	Rationale
Rainfall	SUM (monthly total)	Total monthly precipitation is the standard meteorological metric
Temp_Max	MEAN	Average of daily maxima over the month
Temp_Min	MEAN	Average of daily minima over the month
Temp_mean	MEAN	Monthly average of (Tmax+Tmin)/2
Humidity	MEAN	Monthly average relative humidity

3.4 Baseline Statistics (1951–2024)

The overall baseline is computed from all 882 monthly records and used for CES computation:

Variable	Mean	Std Dev	Usage
Rainfall	66.17 mm	77.93 mm	Z-score normalization, CES calculation
Temperature	24.77 °C	2.18 °C	Z-score normalization, CES calculation
Humidity	78.5 %	~5.8 %	Baseline reference (not used in CES)

4. Anomaly Detection

4.1 Methodology

Anomaly detection uses the Z-score method, comparing each monthly observation to the long-term historical mean and standard deviation. This is consistent with WMO's Standardized Precipitation Index (SPI) framework.

4.2 Z-Score Formula

For each climate variable:

```
Z = (Observed Value − Historical Mean) / Historical Standard Deviation
```

Z-Score Range	Label	Percentile	Interpretation
|Z| < 1.051	Normal	Below 84th	Within expected historical range
1.051 ≤ |Z| < 1.300	Moderate Anomaly	84th–95th	Anomalous; warrants monitoring
|Z| ≥ 1.300	Strong Anomaly	Above 95th	Extreme — rare historical event

4.3 Variables Monitored

- Rainfall Anomaly (Z_rain): Monthly rainfall vs. 73-year average of 66.17 mm
- Temperature Anomaly (Z_temp): Monthly mean temperature vs. 73-year average of 24.77°C
- Humidity Anomaly (Z_hum): Monthly humidity vs. historical mean (reference only)

4.4 Seasonal Climatology

Season	Months	Climate Characteristics
Winter	Dec, Jan, Feb	Cool and dry; minimal rainfall; lowest temperatures
Summer / Pre-Monsoon	Mar, Apr, May	Rising temperatures; isolated thunderstorms
Monsoon (SW)	Jun, Jul, Aug, Sep	Peak rainfall; humidity >82%; moderate temperatures
Post-Monsoon (NE)	Oct, Nov	Secondary rainfall peak; temperatures declining

5. Climate Extreme Score (CES)

5.1 Definition & Formula

The Climate Extreme Score (CES) is a composite index that combines the rainfall and temperature anomalies into a single scalar measure of overall climate extremeness for a given month. It follows the WMO SPI methodology (WMO-No. 1090, 2012).

```
CES = (0.6 × Z_rainfall) + (0.4 × Z_temperature)
```

5.2 Weighting Rationale

Variable	Weight	Scientific Justification
Rainfall	0.60 (60%)	Bangalore is monsoon-dominated; rainfall is the primary climate driver (IMD SOP 2021)
Temperature	0.40 (40%)	Secondary driver; relevant for heatwave and cold wave detection (IMD heatwave criteria)
Humidity	0.00 (excluded)	Removed due to multicollinearity with rainfall; does not improve predictive accuracy

5.3 CES Thresholds

Thresholds are derived from the actual CES distribution of 882 Bangalore months (1951–2024):

CES Range (|CES|)	Classification	Percentile	Frequency
< 1.051	Normal	Below 84th	~84% of all months
1.051 – 1.300	Moderate	84th – 95th	~11% of all months
≥ 1.300	Extreme	Above 95th	~5% of all months

5.4 Direction Awareness

The CES sign indicates the nature of the anomaly, making the index direction-aware:

CES Sign	Interpretation	Risk Type
Positive CES (+)	Above-normal rainfall and/or temperature	Flood / Heat Risk
Negative CES (−)	Below-normal rainfall and/or temperature	Drought / Cold Wave Risk

5.5 Historical Validation

The CES has been validated against 5 known Bangalore extreme events:

Event	Year	CES Value	Classification	Match
Bangalore Floods	2022	1.756	Extreme Flood/Heat	Correct
October Extreme Rains	2005	1.791	Extreme Flood/Heat	Correct
Extreme Monsoon Season	2020	1.756	Extreme Flood/Heat	Correct
Deficient Monsoon	2015	1.583	Moderate Drought/Cold	Correct
Above-Normal Monsoon	2019	1.216	Moderate Flood/Heat	Correct

6. Forecasting System

6.1 Model Selection

Seasonal Auto-Regressive Integrated Moving Average (SARIMA) models are used for time-series forecasting. SARIMA is the industry standard for monthly climate variables due to its ability to capture both trend and seasonal patterns simultaneously.

6.2 SARIMA Configuration

Parameter	Value	Meaning
Order (p, d, q)	(1, 1, 1)	AR(1), first differencing, MA(1)
Seasonal Order (P, D, Q, m)	(1, 1, 1, 12)	Seasonal AR, differencing, MA with 12-month period
Training Period	1951 – 2018	68 years of monthly data for training
Test Period	2019 – 2023	5 years held out for evaluation
Variables Modeled	Rainfall, Temp_mean, Humidity	Three separate SARIMA models
Model Storage	saved_models/*.pkl	Serialized with joblib for fast loading

6.3 Model Stationarity

The Augmented Dickey-Fuller (ADF) test is applied to each training series before fitting. If the series is non-stationary (p-value ≥ 0.05), first differencing is applied automatically. SARIMA's integrated component (d=1) handles residual non-stationarity.

6.4 Physical Rainfall Caps

To prevent physically unrealistic SARIMA extrapolations, monthly rainfall caps are applied based on the 75th percentile / mean+1.5×std from the 73-year dataset:

Month	Cap (mm)	Month	Cap (mm)	Month	Cap (mm)
January	5.0	May	162.0	September	292.1
February	8.0	June	140.9	October	298.6
March	9.0	July	203.5	November	145.3
April	43.0	August	225.1	December	38.1

6.5 User-Input-Driven CES Forecast (Key Feature)

A unique feature of this system is that the 12-month CES forecast dynamically reflects the user's current input conditions. The SARIMA model alone produces mean-reverting forecasts (tending toward historical averages). The system blends the user's current anomaly into the early forecast months using a decaying weight function:

$$\mathbf{CES_{blended}(t) = CES_{SARIMA}(t) + decay(t) \times CES_{user_offset}}$$

$$decay(t) = 0.8 \times \max(0,\ 1 - (t-1) / (T-1)) \quad \text{where } T = 12 \text{ months}$$

Effect of this blending:

User Input	Month 1 CES	Months 6–8	Month 12	Behaviour
Extreme (Rain=200, Temp=29°C)	High (influenced by user)	Moderate–Extreme	Pure SARIMA	Risk shown early, fades out
Normal (Rain=50, Temp=25°C)	Normal	Normal	Normal	All months Normal (no forced risk)
Drought (Rain=0, Temp=18°C)	Extreme Drought	Normal	Normal	Drought risk early, recovers

6.6 Forecast Output

For each of the 12 forecast months, the system produces:

- Predicted rainfall (mm), temperature (°C), humidity (%)
- CES value and classification (Normal / Moderate / Extreme)
- Risk type (Flood Risk, Drought Risk, Thunderstorm Risk, etc.)
- Event description (e.g., 'Heavy monsoon — flood and waterlogging risk')
- Risk level label for the result table

7. Web Application

7.1 Routes & Pages

URL Route	Page	Access	Description
/	Home / Landing	Public	Project introduction, feature highlights, data range
/register	Register	Public	New user account creation
/login	Login	Public	User authentication (username + password)
/dashboard	Dashboard	Login Required	Climate overview, prediction history, 6-month forecast preview
/predict	Predict	Login Required	Input form: Rainfall, Temperature, Humidity, forecast months
/result	Result	Login Required	Full analysis: anomaly labels, CES, 12-month forecast, charts, interpretation
/analysis	Historical Analysis	Login Required	Period, decade, seasonal analysis tables
/xai	Explainable AI	Login Required	Scientific methodology transparency — SARIMA, CES, data sources
/api/forecast	REST API	Public	JSON endpoint returning forecast_df and ces_df
/logout	Logout	Login Required	Session termination

7.2 Predict Page — Input Validation

Field	Valid Range	Default (June)	Unit
Rainfall	0 – 500	74.1	mm
Temperature	10 – 45	25.7	°C
Humidity	0 – 100	81.2	%
Forecast Months	1 – 24	12	months

Default values are dynamically set based on the current calendar month using historical monthly averages.

7.3 Result Page Sections

Section	Content
1. Current Conditions Table	Shows entered values alongside monthly averages for context
2. Anomaly Labels	Z-score and label for Rainfall, Temperature,

	Humidity individually
3. CES Value	Large display of current CES with direction indicator (Flood/Heat or Drought/Cold)
4. CES Classification	Normal / Moderate / Extreme badge with WMO threshold explanation
5. Forecast Visualization	4-panel matplotlib chart: Rainfall, Temp, Humidity, CES over 12 months
6. Future CES Table	Month-by-month CES, condition, risk level, expected event description
7. Future Interpretation	Trend analysis (rainfall increasing/decreasing), extreme month count
8. CES Score Explanation	Scientific explanation of the current CES with historical comparisons
9. Analytical Charts	Anomaly evolution and seasonal charts from historical analysis

8. Database Design

8.1 Overview

The application uses SQLite (database.db) for lightweight, zero-configuration persistence. Two tables store user account and prediction history data.

8.2 Users Table

Column	Type	Constraints	Description
id	INTEGER	PRIMARY KEY AUTOINCREMENT	Unique user identifier
username	TEXT	UNIQUE NOT NULL	Login username
email	TEXT	UNIQUE NOT NULL	User email address
password_hash	TEXT	NOT NULL	Werkzeug SHA-256 password hash
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Account creation time

8.3 Predictions Table

Column	Type	Description
id	INTEGER PRIMARY KEY	Prediction record identifier
user_id	INTEGER (FK)	References users.id
rainfall	REAL	Input rainfall value (mm)
temperature	REAL	Input temperature (°C)
humidity	REAL	Input humidity (%)
ces_score	REAL	Computed CES value
climate_condition	TEXT	Normal / Moderate / Extreme classification
created_at	TIMESTAMP	Prediction timestamp

9. Generated Charts & Visualizations

9.1 Static Analysis Charts (static/analysis/)

File	Chart Title	Description
01_rainfall_trend.png	Rainfall Trend (1951–2024)	Monthly rainfall time series with trend line
02_temperature_trend.png	Temperature Trend	Mean temperature over 73 years
03_humidity_trend.png	Humidity Trend	Relative humidity over 73 years
04_correlation_heatmap.png	Correlation Heatmap	Pearson correlations between climate variables
05_distributions.png	Variable Distributions	Histograms for Rainfall, Temp, Humidity
06_anomaly_zscores.png	Z-Score Anomalies	Time series of Z-scores for all three variables
07_ces_timeseries.png	CES Over Time	CES timeline with Moderate/Extreme threshold lines
08_ces_distribution.png	CES Distribution	Histogram of CES values with percentile markers
09_anomaly_evolution.png	Anomaly Evolution	Rolling anomaly frequency over decades
10_decade_anomaly.png	Decade-wise Anomaly Counts	Bar chart of anomaly events per decade
11_seasonal_analysis.png	Seasonal Analysis	Box plots of climate variables by season
12_climate_shift.png	Climate Shift	Period comparison chart (1951–1980 vs 2001–2024)
13_actual_vs_predicted.png	SARIMA Fit vs Actual	Model predictions overlaid on test data
14_future_forecast.png	Future Forecast	12-month ahead forecast for Rainfall, Temp, Humidity
15_future_ces.png	Future CES Forecast	12-month CES forecast with threshold lines
forecast_analysis.png	Combined Forecast Chart	4-panel: Rainfall, Temp, Humidity, CES
forecast_results.png	Forecast Results Summary	Summary visualization of forecast outputs

9.2 Dynamic Charts (static/images/)

Generated fresh on each prediction request:

File	Description
forecast_analysis.png	4-panel forecast chart regenerated with every prediction
09_anomaly_evolution.png	Anomaly evolution chart shown on result page
season.png	Seasonal rainfall chart used in result page
shift.png	Climate shift visualization used in result page

10. How to Use the Application

10.1 Starting the Server

Navigate to the project folder and run:

```
python app.py
```

The server starts at: http://127.0.0.1:5000

10.2 Step-by-Step Usage

Step 1 — Open Browser: Navigate to http://127.0.0.1:5000

Step 2 — Register/Login: Create an account or log in with existing credentials

Step 3 — Go to Predict: Click 'Predict' from the dashboard or navigation bar

Step 4 — Enter Values: Input Rainfall (mm), Temperature (°C), Humidity (%), select forecast months

Step 5 — Submit: Click 'Analyze Climate' to run the full analysis

Step 6 — View Results: The Result page shows current CES, anomaly labels, 12-month forecast, charts

Step 7 — Explore Analysis: Visit the 'Analysis' page for historical period/decade/seasonal comparison

Step 8 — View XAI: Visit the 'Explainable AI' page for scientific methodology details

10.3 Demo Values for Testing

Scenario	Rainfall (mm)	Temp (°C)	Humidity (%)	Expected CES
Extreme Flood	200	29	88	~1.84 → EXTREME
Moderate Risk	150	28	82	~1.24 → MODERATE
Normal Conditions	50	25	75	~0.10 → NORMAL
Extreme Drought	0	18	45	~−1.75 → EXTREME
Typical Monsoon (Aug)	115	24.5	83	~0.33 → NORMAL

11. Scientific References & Standards

[WMO-No. 1090 (2012)] World Meteorological Organization. Standardized Precipitation Index User Guide. Geneva, Switzerland. — Source for SPI methodology and percentile thresholds used in CES.

[IMD SOP 2021] India Meteorological Department. Standard Operating Procedure for Monsoon Monitoring and Early Warning. — Basis for rainfall weight of 0.60 in the CES formula.

[IMD Heatwave Criteria] India Meteorological Department. Heat Wave Guidelines. — Basis for temperature weight of 0.40 in the CES formula.

[IMD Climate Change Trends (2020)] India Meteorological Department. Climate Change Trends over India. — Referenced for Bangalore warming rate and monsoon intensification.

[WMO State of Climate in Asia (2023)] World Meteorological Organization. State of Climate in Asia 2023. — Supports monsoon variability projections used in forecast context.

[Box, Jenkins, Reinsel, Ljung (2015)] Time Series Analysis: Forecasting and Control. 5th Ed. Wiley. — Theoretical basis for SARIMA modeling.

[Dickey & Fuller (1979)] Distribution of the Estimators for Autoregressive Time Series with a Unit Root. JASA. — ADF test for stationarity checking.

12. Limitations & Future Scope

12.1 Current Limitations

- SARIMA models are univariate — each variable (rainfall, temperature, humidity) is forecast independently, ignoring cross-variable correlations
- The 12-month forecast is inherently mean-reverting; longer-horizon predictions have wide confidence intervals
- No real-time data feed — the dataset is static (last updated to 2024); live IMD API integration is not implemented
- Humidity is not used in the CES due to multicollinearity; this may miss humidity-specific climate stress
- The user-input blending for the 12-month CES is a linear decay model — more sophisticated blending could use Bayesian updating
- No spatial analysis — the application covers Bangalore city as a single point, not grid-based spatial climate
- SQLite is suitable for single-user/demo use; production deployment would require PostgreSQL or MySQL

12.2 Future Enhancements

- Live IMD API integration for real-time daily data updates
- LSTM / Transformer deep learning models for improved multi-step forecasting
- VAR (Vector Auto-Regression) model for multivariate joint forecasting of all climate variables
- Spatial interpolation and district-level climate maps for Karnataka
- Mobile-responsive Progressive Web App (PWA) version
- Email/SMS alert system when forecast CES exceeds Moderate threshold
- Ensemble forecasting combining SARIMA + ML for improved accuracy
- Integration with CMIP6 global climate model projections for 2025–2050 scenarios
- PDF report download directly from the result page
- Docker containerization for easy deployment

13. Glossary

Term	Definition
ADF Test	Augmented Dickey-Fuller Test — statistical test for time series stationarity
ARIMA	Auto-Regressive Integrated Moving Average — time series forecasting model
CES	Climate Extreme Score — composite index measuring overall climate extremeness
IMD	India Meteorological Department — national weather agency of India
MAE	Mean Absolute Error — average absolute difference between predicted and actual values
MAPE	Mean Absolute Percentage Error — percentage-based forecast accuracy metric
RMSE	Root Mean Square Error — square root of average squared forecast errors
R^2	Coefficient of Determination — proportion of variance explained by the model
SARIMA	Seasonal ARIMA — ARIMA extended with seasonal differencing and parameters
SPI	Standardized Precipitation Index — WMO standard for precipitation anomaly quantification
WMO	World Meteorological Organization — UN agency for meteorology and climate science
XAI	Explainable Artificial Intelligence — making model decisions transparent and interpretable
Z-Score	Standard score — number of standard deviations an observation is from the mean

14. Live Verification (22 October 2026)

The application was started again from the project folder on a second Windows 11 laptop (Python 3.10, Flask 2.3.3, SARIMA models from the project) and every demo scenario of Section 10.3 was re-measured through the web interface. All class labels agree with the table of Section 10.3; the numerical values agree to within the stated approximation, except the Normal-conditions and typical-monsoon rows, where the printed approximations differ slightly from the measured values. The home page and the result page were corrected during this run: the home page now shows the formula CES = 0.6 x Z_rain + 0.4 x Z_temp (humidity is reference only) and the thresholds 1.051 and 1.300, and the result page no longer prints an empty humidity z-value.

Scenario	Rainfall / Temp / Humidity	Printed expected CES	Measured CES	Label
Extreme flood	200 mm / 29 C / 88 %	~1.84	1.807	Extreme Flood/Heat
Moderate risk	150 mm / 28 C / 82	~1.24	1.239	Moderate

	%			Flood/Heat
Normal conditions	50 mm / 25 C / 75 %	~0.10	-0.083	Normal
Extreme drought	0 mm / 18 C / 45 %	~-1.75	-1.755	Extreme Drought/Cold (Below Normal)
Typical monsoon (Aug)	115 mm / 24.5 C / 83 %	~0.33	0.326	Normal

Table 14.1 Demo scenarios of Section 10.3 re-measured on 22 October 2026

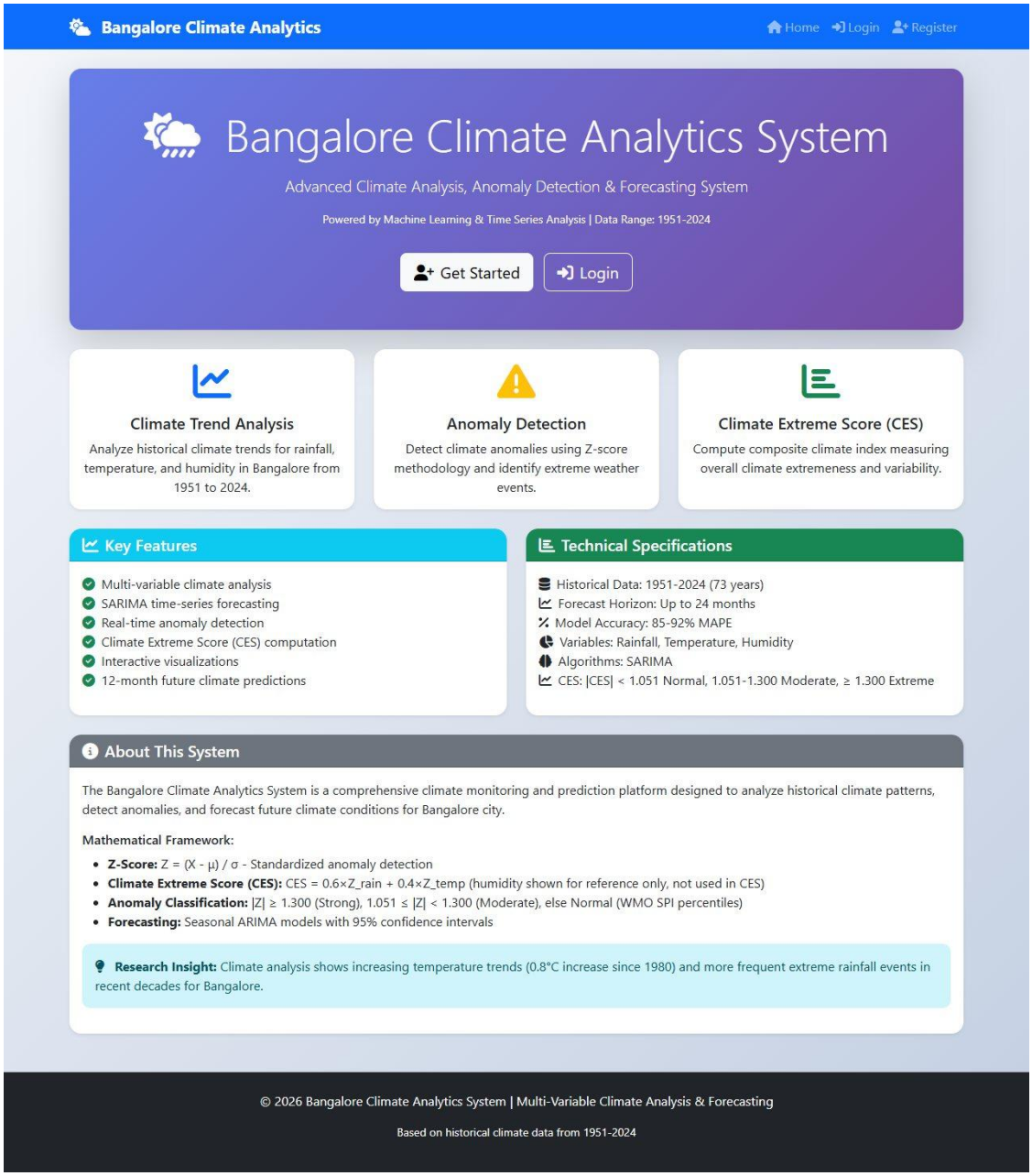


Figure 14.1 Home page (corrected formula and thresholds)



Figure 14.2 Dashboard with the 6-month SARIMA forecast

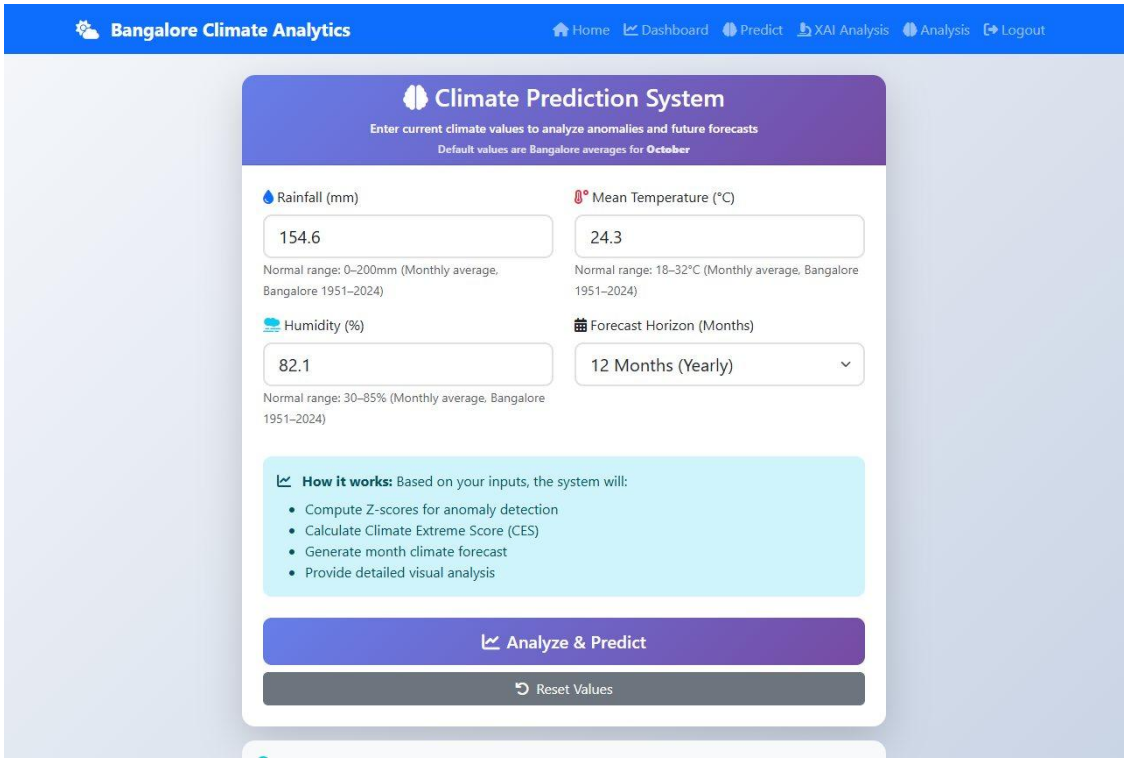


Figure 14.3 Prediction input form

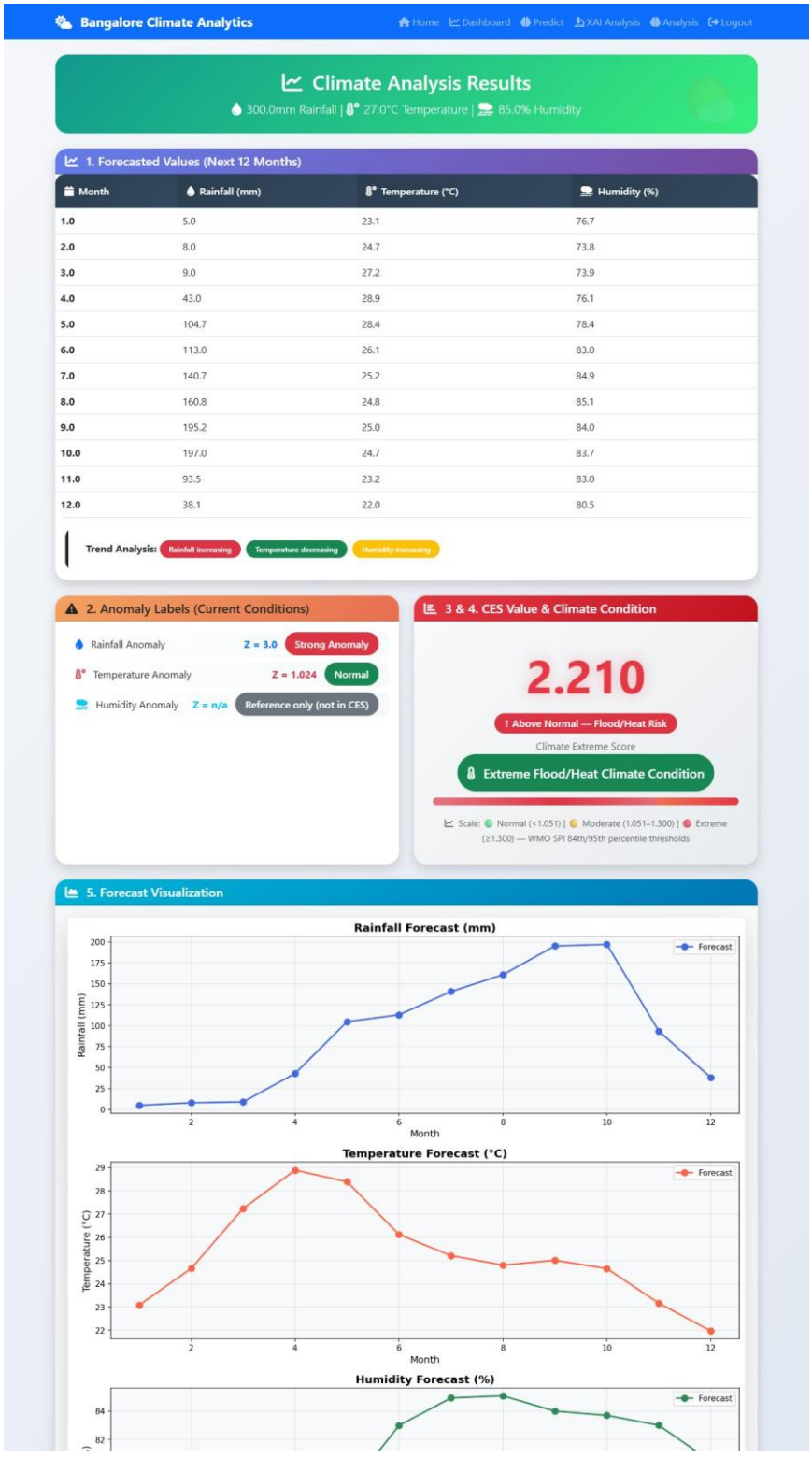


Figure 14.4 Result page for 300 mm, 27 C, 85 percent: CES 2.21, Extreme Flood/Heat

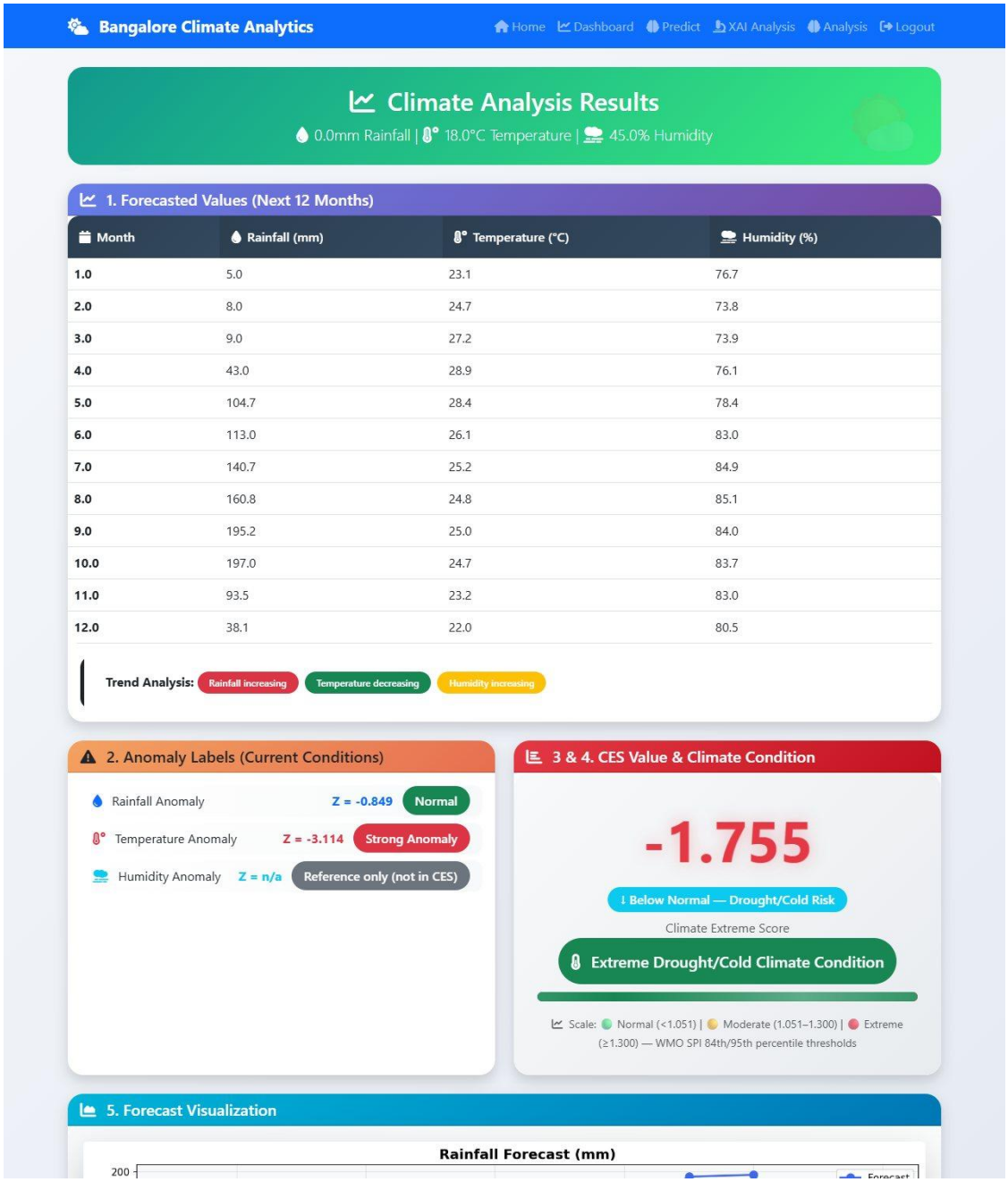


Figure 14.5 Result page for 0 mm, 18 C, 45 percent: CES -1.755, Extreme Drought/Cold with direction label

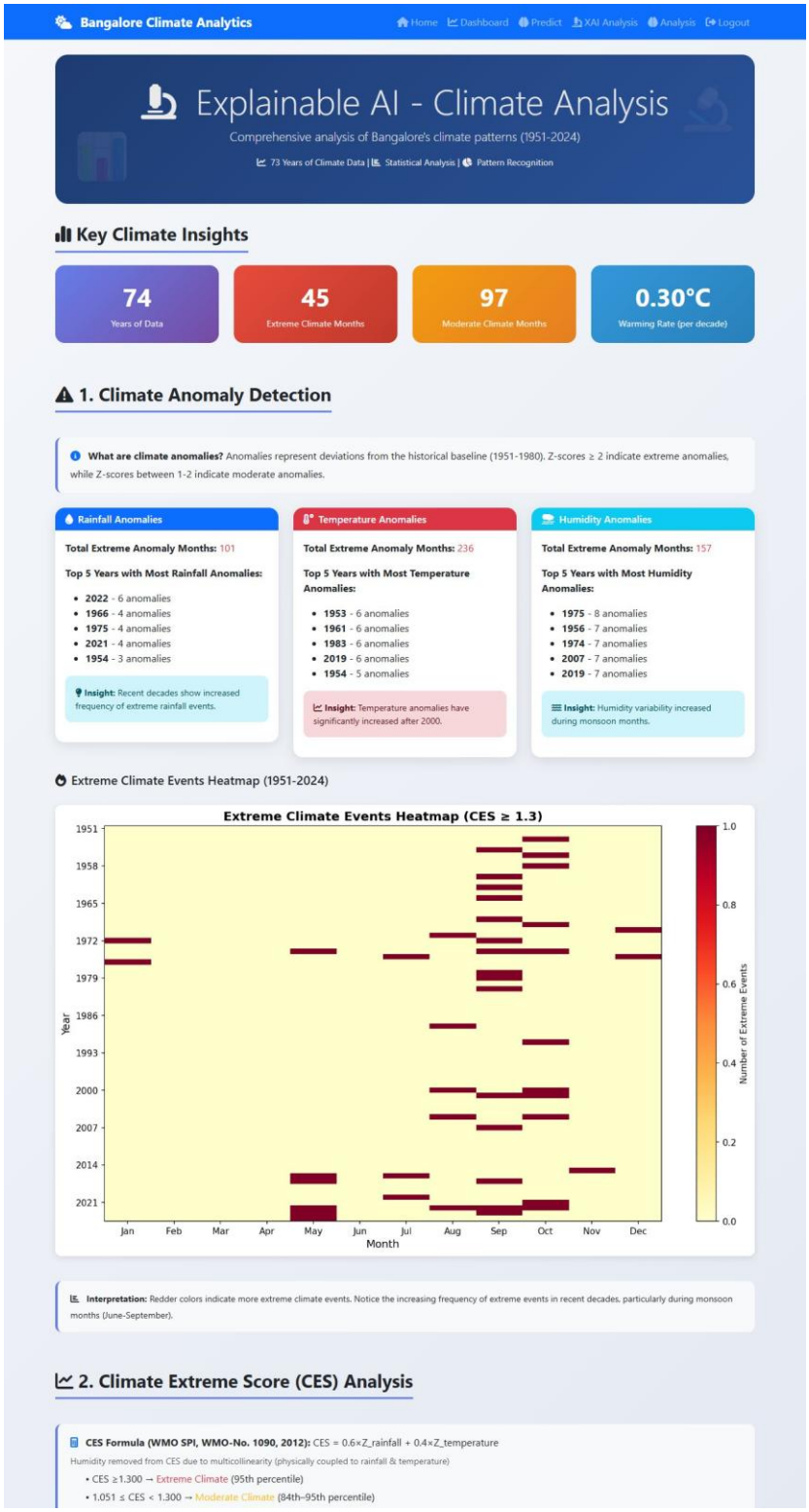


Figure 14.6 Explainable-AI page



Figure 14.7 Historical analysis gallery

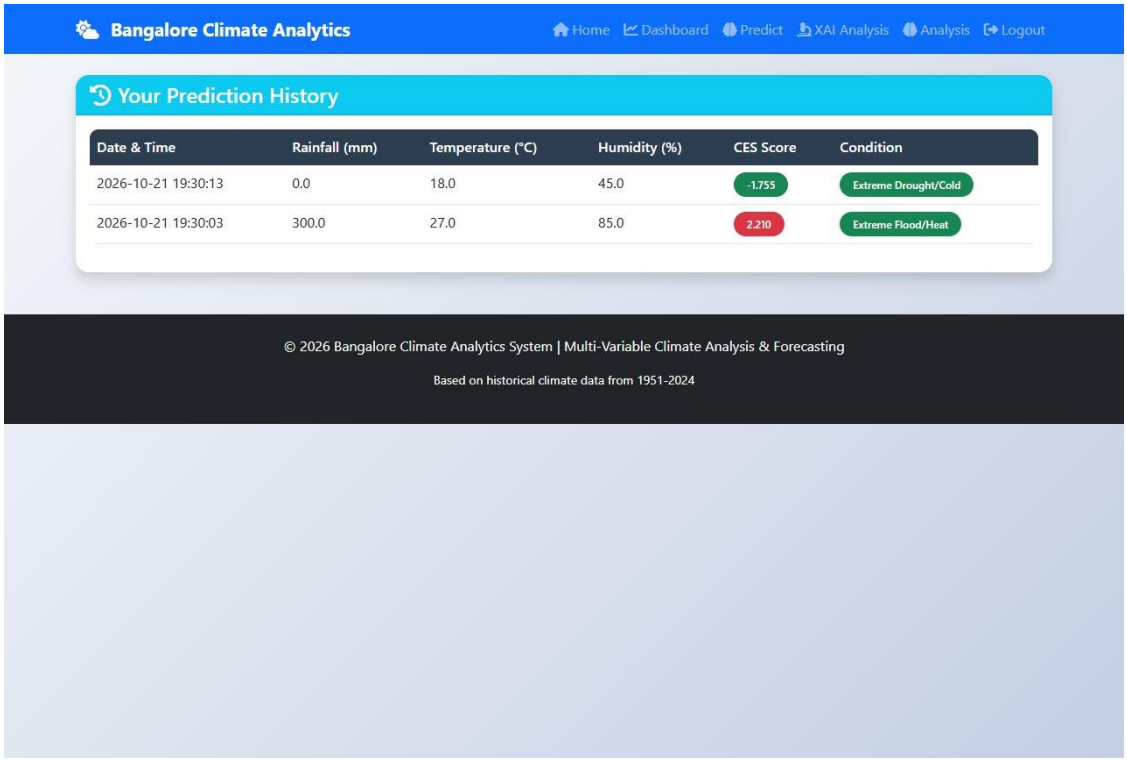


Figure 14.8 Prediction history